User Manual RFS-4220T

Signal Generator

WARRANTY

All Berkeley Nucleonics (BNC) instruments are covered by a 1 year return-to-manufacturer (90 days high current/high voltage pulsers) repair warranty for parts and labor from the date shipment against defective materials and/or workmanship provided the product has been subjected to proper use, handling and storage. Berkeley Nucleonics shall not be responsible for defects caused by natural wear and tear, willful damage, negligence, force majeure, and abnormal working conditions or failure to follow Berkeley Nucleonics spoken or written guidelines as to the storage, installation, commissioning, use or maintenance of the products or (if there are none) good trade practice; or the customer alters or repairs such goods without written consent from Berkeley Nucleonics. Opening the instrument and/or breaking warranty seals voids the warranty. Service plans are subject to the same warranty policy regarding use.

IMPORTANT! PLEASE READ CAREFULLY

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General Warranty

BERKELEY NUCLEONICS warrants that the product will be free from defects in materials and workmanship for 1 year from the date of purchase of the product by the original purchaser from the BERKELEY NUCLEONICS Company. The warranty period of accessories such as probe, battery, adapter is 12 months. This warranty only applies to the original purchaser and is not transferable to the third party.

If the product proves defective during the warranty period, BERKELEY NUCLEONICS either will repair the defective product without charge for parts and labor or will provide a replacement in exchange for the defective product. Parts, modules and replacement products used by BERKELEY NUCLEONICS for warranty work may be new or reconditioned like new performance. All replaced parts, modules and products become the property of BERKELEY NUCLEONICS.

To obtain service under this warranty, Customer must notify BERKELEY NUCLEONICS of the defect before the expiration of the warranty period. Customer shall be responsible for packaging and shipping the defective product to the service center designated by BERKELEY NUCLEONICS, and with a copy of customer proof of purchase.

This warranty shall not apply to any defect, failure or damage caused by improper use or improper or inadequate maintenance and care. BERKELEY NUCLEONICS shall not be obligated to furnish service under this warranty a) to repair damage resulting from attempts by personnel other than BERKELEY NUCLEONICS representatives to install, repair or service the product; b) to repair damage resulting from improper use or connection to incompatible equipment; c) to repair any damage or malfunction caused by the use of non-BERKELEY NUCLEONICS supplies; or d) to service a product that has been modified or integrated with other products when the effect of such modification or integration increases the time or difficulty of servicing the product.

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GENERAL REMARKS

The RFS-4220T is a compact, high-performance RF and microwave signal generator designed for lab testing, automated systems, production environments, and RF development. Covering 0.1 GHz to 20 GHz calibrated, settable to 22 GHz, the instrument provides precise frequency control, fine power adjustment, low phase noise, excellent spectral purity, and flexible local or remote operation.

QUICK START GUIDE

Power your device using the provided cable and hold the power button to power on. The unit will automatically work in touchscreen mode, while the front buttons may take a minute to connect. Once connected, the status text will update from “Control panel loading…” to “Control panel connected”

Reference Input (Optional)

- Connect an external 10 MHz reference to the REF IN port if preferred. Otherwise, the device will default to its internal OCXO source.

Sweep Trigger Setup (Optional)

- Connect an external 5 V TTL pulse to the TRIG connector to initiate the saved sweep (or single step if configured) at the rising edge.

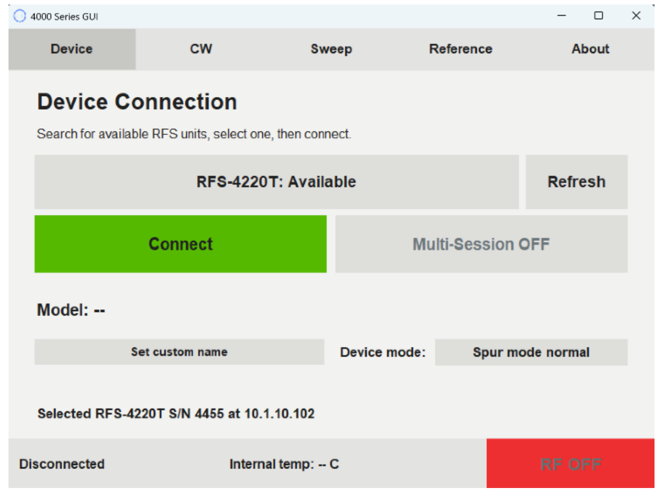
External Software (Optional)

- Download the 4000_Series_GUI_Zip_Installer.zip from the BNC website (berkeleynucleonics.com). Right-click the downloaded ZIP file and select Extract All. Open the extracted folder and double-click Install_4000_Series_GUI.cmd.

GUI OVERVIEW

Device Page (external software)

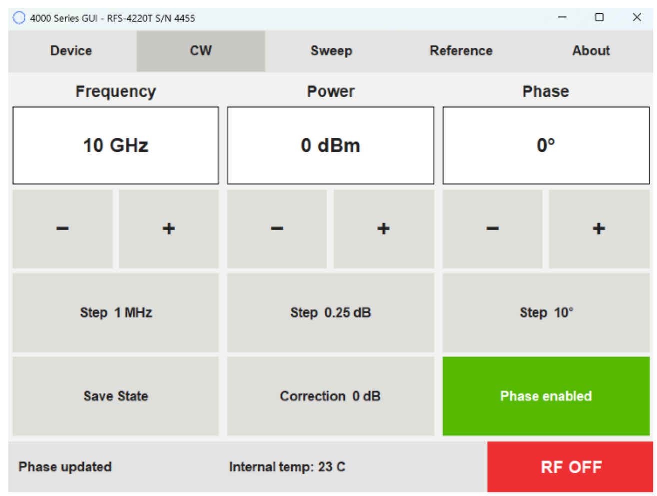
- **Refresh** finds all available RFS units on the local network
- **Multi-session** toggle allows multiple computers to control the same device
- **Custom names** are stored internally on the device
- **Device mode** dropdown controls how the device handles unwanted spurs



1. Device Page

CW Page

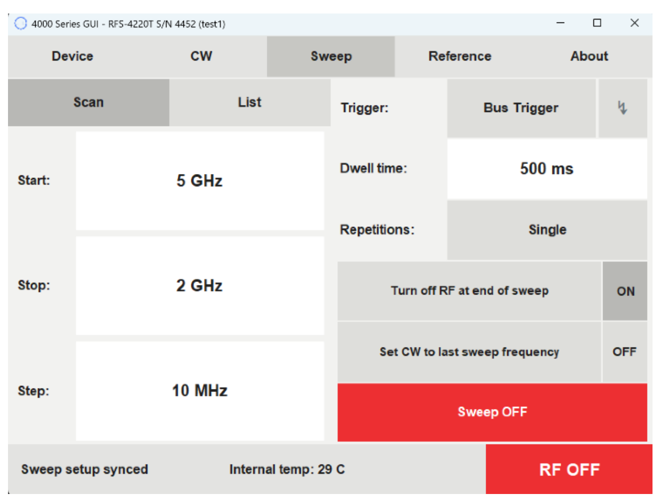
- Set **frequency**, **power**, and **phase** by tapping input boxes or using **step** and **+/-** buttons
- **Save state** before turning unit off to restore frequency on startup
- Set **correction** to compensate for external power loss or gain
- Recommended to **disable phase** when not needed to reduce noise



2. CW Page

Sweep Page (Scan)

- Set sweep **start**, **stop**, and **step** using numpad
- **Trigger** dropdown determines sweep initiator
  - **Internal:** sweep starts on "Sweep OFF" press
  - **Bus:** sweep steps incrementally with each press of " ϟ "button
  - **External full list:** external trigger initiates entire sweep
  - **External single step [dwell]:** external trigger increments one step and outputs for duration of dwell time
  - **External single step [inf]:** external trigger increments one step and outputs until next trigger

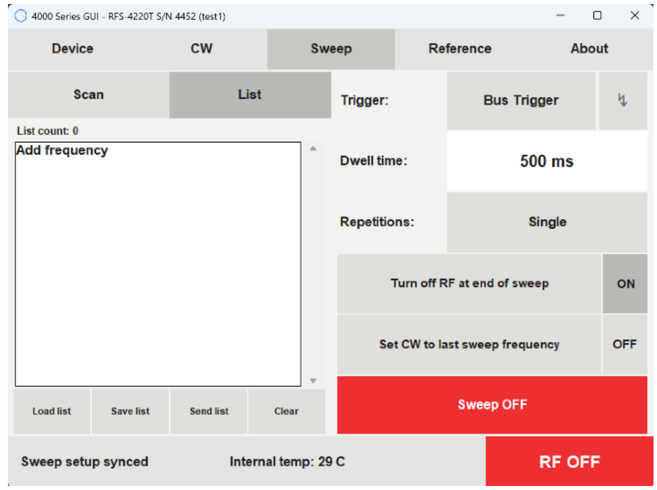


3. Sweep Page (Frequency Mode)

Sweep Page (List)

- Tap anywhere to **add frequency**
- Select a frequency to delete or edit
- **Load list** from internal storage, or from computer if using computer application
- **Save list** to internal storage, or to computer if using computer application

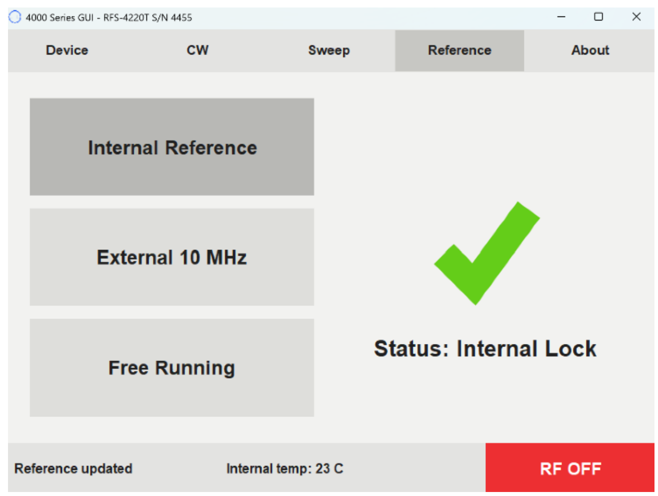
  This allows lists to be easily shared between computers and units!
- **Send list** to arm configuration for trigger
- Set **dwell time** to determine step duration
- Toggle **repetitions** between single and infinite



4. Sweep Page (List Mode)

Reference Page

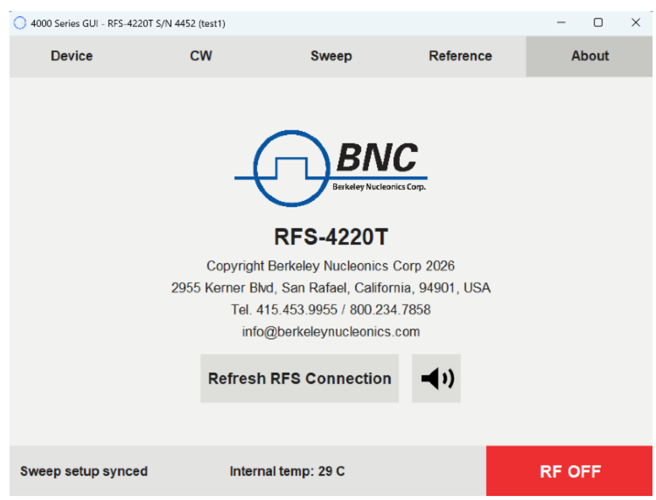
- **Internal reference** uses internal OCXO clock
- **External reference** connects to externally supplied 10 MHz clock if present. If one is not detected, system defaults to internal
- **Free running** uses no reference clock
- **Status** displays lock status. If reference is unlocked, refresh RFS connection on About Page



5. Reference Page

About Page

- Contains BNC address and contact information
- **Refresh RFS Connection** when status shows reference is unlocked or when status shows “RF STATE UNKNOWN
- **Speaker** button mutes/unmutes unit



6. About Page

SCPI COMMAND LIST

Command	Parameters	Comment
***UNITNAME**	<string>	Set a unique name in flash memory
***RST**	-	Reset unit
***DISPLAY**	ON, OFF	Power ON or OFF the display
***BUZZER**	ON, OFF	Mute the buzzer
***SAVESTATE**	-	Save frequency & attenuation as boot defaults
OUTP:STAT	1, 0	Turn on or off the RF output
FREQ:CW	1MHz to 22GHz	Set output Frequency in Hz
FREQ:START	1MHz to 22GHz	Sweep start frequency in Hz
FREQ:STOP	1MHz to 22GHz	Sweep stop frequency in Hz
POWER	-45dBm to +15dBm	Set output RF level in dBm
INIT:IMM	-	Start the sweep now
INIT:CONT	-	Sweep continuous mode or single
ABORT	-	Stop the sweep now
LIST:ADD	-	Add a point to the end of the sweep list
LIST:CLEAR	-	Clear the working list
LIST:DIR	UP, DOWN	Sweep direction
SWE:MODE	OFF, SCAN, LIST	Selects sweep mode
SWE:DWELL	5 to 10000 ms	Sweep dwell time
SWE:POINTS	<integer>	Sweep point count
SWE:RESET	-	Return to the start of the point list (does not clear memory)
SWE:ENDRFO	ON, OFF	Determines output state at the end of a sweep
INIT:STEPMODE	0, 1, 2	Trigger activates the entire sweep; trigger steps one point, outputs for the dwell time, then stops; trigger steps one point and outputs indefinitely until next trigger
SYSREF	AUTO, FREE, INT, EXT	Set reference to auto detect, free running, internal, or external 10MHz

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